

- Zero-knowledge vault
- Per-employee encryption
- Immutable audit trail
- DPDP Act 2023 compliant

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Personal Repository And Networked Authentications

- Cryptographically Yours — architecture, not policy
- Verified, Not Just Stored — AI-extracted, employer-pushed
- Follows You, Forever — one URL, every employer, for life



prana.in/vault/you

One URL. Permanent. Yours.

- Salary Slip · Apr 2026 · Employer A
- Form 16 · FY 2025-26 · Employer B
- Offer Letter · 2022 · Employer C

 PROBLEM STATEMENT

India's workforce has

Crores of workers change jobs and accumulate salary slips, Form 16s, offer letters — scattered across email, WhatsApp, and company intranets they can never access again once they leave.



No permanent home

Documents live with employers. Access cut off within 24 hours of resignation — sometimes immediately on exit.



No verification layer

Banks and background firms cannot verify salary claims without slow, expensive, manual HR calls.



No compliance posture

DPDP Act 2023 creates ₹50Cr exposure for mishandled employee data. Most HR systems have zero document-level audit trail.



No trust architecture

HR admins can open, copy, or share any employee document. Only policies — no structural controls that survive credential compromise.

Maximum DPDP Act 2023 penalty per violation. Most organisations have zero structural controls at the document layer.

Existing platforms give employees cryptographic ownership of their own career documents.

Built so that misuse is structurally impossible — not just policy-restricted. Architecture is the control.

 PLATFORM DEFINITION

A permanent, portable document identity

PRANA is a multitenant SaaS platform that gives every employee a single, permanent, portable document identity — one URL that persists across employers, survives job changes, and accumulates documents from every organisation that participates in the programme.



PROMISE 1

Once a document reaches your vault, your employer cannot touch it. Not open it, not delete it, not share it. This is not a policy. It is how the mathematics works.



PROMISE 2

Every document was pushed by your employer, extracted by AI, and timestamped immutably. When you share a salary slip, the system proves it is real.



PROMISE 3

One vault. Every employer. Your entire career automatically assembled — whether employed, between jobs, or 10 employers in. The URL never changes.

One Employee • One URL • Forever

Documents flow in from every participating employer. Verification via cryptographic proof — not a phone call to HR. The vault is permanently yours from the first document.

👥 FOUR STAKEHOLDERS · ONE PLATFORM · ALIGNED INCENTIVES

Every role gets a — not just access



PRANA

Permanent · Portable ·
Cryptographic · DPDP-native

One URL per employee

Per-employee DEK
encryption

Employer push → vault
ownership transfer

AI pipeline · 6 stages

53 Temporal workflows

All 5 DPDP rights built in



EMPLOYEE

Stop chasing HR for documents you've already earned

Your vault is yours — across every employer, for life. You control who sees what, for how long.

- **2ms** — PAN destroyed forever
- **₹0** — prove salary to any bank
- **0** — employer accesses post-exit



CHRO / HR

Stop scrambling when auditors arrive

Every obligation automatic, auditable, provable. Digests generated without manual steps.

- **100%** — compliance timestamped at push
- **1-click** — audit export any period
- **0** — manual steps for monthly digest



CFO / FINANCE

Stop taking your CHRO's word for it

Payroll analytics from verified documents. Anomalies surfaced automatically, not at year-end.

- **30 min** — cohort before aggregate shown
- **₹50Cr** — penalty structurally impossible
- **0** — self-reported inputs in analytics



INFOSEC / DPO

Not policy. Architecture.

Controls fail when credentials are compromised. PRANA prevents misuse regardless of who has the keys.

- **1 person** — exposed per DEK compromise
- **0** — plaintext PAN in any DB row
- **2ms** — PAN lifetime in memory

 EMPLOYEE VALUE PROPOSITION

Your entire career

The mobile app is the employee's window into PRANA. Documents accumulate automatically. Sharing is controlled, watermarked, time-limited. No employer can reach the vault after routing.



Permanent vault URL — one identity, every employer

The vault URL never changes across job moves. New employer documents flow in automatically when they join PRANA.



Named, time-limited, watermarked sharing

Generate share link for a bank: set recipient name, expiry, OTP auth. Watermark applied server-side — cannot be removed. Revocable at any time.



Full activity visibility — every access, every actor

Actor, timestamp, channel logged — visible in your activity feed. No black boxes. PRANA cannot hide access from you.



All 5 DPDP rights — Access, Erasure, Correction, Nomination, Consent

One tap to exercise any right. Tracked to completion via Temporal workflow. No chasing the DPO by email.

Employer accesses to your documents after exit — structurally enforced, not policy-restricted

PAN lifetime in memory before cryptographic destruction — never stored plaintext in any database row

Cost to prove salary to any bank or lender — cryptographic proof replaces the HR phone call entirely

CHRO VALUE PROPOSITION

Stop scrambling when

Every document obligation timestamped at the moment of push. Compliance is automatic — not a quarterly scramble. PRANA gives CHRO evidence, not just a summary.

	Auditor requests 3 years of salary slip proof Every document cryptographically timestamped at push. Full provenance chain included.	1-click export ⚡
	Monthly ESIC / PF digest preparation DigestWorkflow runs on schedule. CHRO receives the finished report automatically — no spreadsheets.	0 manual steps ✓
	Vault completeness tracking by department Completeness tracked against every employee's document obligations. Gaps surface automatically.	Real-time dashboard 
	Exception queue — AI cannot resolve identity OA-Admin resolves via portal. Temporal workflow tracks SLA. Escalates to Portal Admin if breached.	SLA-bound ⌚
	Employee DPDP erasure request ErasureWorkflow handles PII deletion, S3 purge, DPDP certificate. 7-year audit log retained.	Auto 30-day ⚖️

Compliance timestamped at document push. No retroactive certification needed.

Audit export — any period, any department, any document type with full provenance.

Manual steps for monthly digest. CHRO receives the finished report automatically.

 CFO VALUE PROPOSITION

Stop taking your CHRO's

Payroll compliance and attrition cost derived from verified documents — not self-reported data. Anomalies surface automatically, not at year-end.



Verified Payroll Compliance

Every salary slip and Form 16 is employer-verified before entering analytics. No self-reported numbers. No manual reconciliation.



Anomaly Detection

Payroll anomalies surfaced with document-level evidence. CFO acknowledges via portal — logged to audit trail instantly.



Cohort Privacy by Default

No aggregate insight shown until cohort meets minimum 30 people. Prevents reverse-engineering individual salaries.



DPDP Penalty Elimination

₹50Cr max DPDP fine structurally impossible when architecture enforces the controls. CFO does not rely on policy.



7-Year Immutable Audit Trail

Every access, acknowledgement, and export written to immutable log. Court-admissible. YugabyteDB hot + Iceberg cold.



Attrition Cost from Documents

Career insights from AI-extracted document data — not HR surveys. Separation costs tracked against actual tenures and grades.

Minimum cohort before any salary aggregate is surfaced. Structural privacy — not a setting.

Max DPDP penalty — made structurally impossible by the encryption architecture.

Self-reported inputs in CFO analytics. Every data point verified and document-sourced.

INFOSEC VALUE PROPOSITION

Not policy. Architecture.

Controls fail when credentials are compromised. PRANA's architecture prevents misuse regardless of who has the keys.

Person's data exposed per DEK compromise — maximum blast radius

Plaintext PAN in any database row, log, or cache — ever

PAN lifetime in memory before cryptographic destruction

Full IP in document access log — employee sees city only, CISO sees full IP

Force-logout: revoke any session instantly — effective within milliseconds via Redis revocation

Login feed: outcome filter, flagged sessions, foreign IP detection, bulk-access anomaly

CISO DASHBOARD

- Account lock dashboard — POLICY_LOCK, BULK_ACCESS_ANOMALY, auto-unlock timer
- Flagged access review — anomalous document access marked and escalatable
- Elevation audit — every elevated OA session: reason, duration, all actions taken
- 7-year immutable audit log — court-admissible, DPDP Act certified
- One-click DPB escalation if employee grievance unresolved at 30 days

 7 LAYERS OF STRUCTURAL TRUST — EVERY CLAIM BACKED BY ARCHITECTURE

Architecture is the policy.

01	 Vault Ownership Transfer	STRUCTURAL	Employer cannot open, download, share, or delete. Ownership transferred structurally at the moment of routing — not configurable by any admin.
02	 One Person · One Key	CRYPTOGRAPHIC	Each employee has their own Document Encryption Key. A colleague's key cannot decrypt yours. AWS KMS envelope encryption enforces this.
03	 Zero-Knowledge PAN	ZERO-KNOWLEDGE	PAN → HMAC-SHA256 in 2ms. Plaintext destroyed. No database row links PAN to name. Cross-tenant dedup via HMAC token only — no PAN ever stored.
04	 Consent-Gated Sharing	CONSENT-GATED	Only the employee creates share tokens. Named, time-limited, revocable, watermarked. No employer or admin can share on the employee's behalf.
05	 PostgreSQL RLS	RLS ENFORCED	Portal Admin has zero SELECT on document rows. Row-Level Security enforces this at the database query layer — the query is refused, not hidden by UI.
06	 Full Transparency	TRANSPARENT	Every access — actor, timestamp, channel — is in the employee's activity log. No black boxes. No hidden reads. PRANA cannot hide access from you.
07	 Portable Vault	PORTABLE	Your vault follows you — not your employer. When you leave, your vault stays yours. New employer pushes to the same permanent URL. Career spans employers.

🔒 DATA PROTECTION MODEL

Envelope encryption.

Three independent encryption layers protect every piece of employee data. A breach at any one layer does not expose data protected by the others. The mathematics is the control.

	PAN Token — Cross-Tenant Deduplication Key HMAC-SHA256(PAN, platform_secret) · 2ms lifetime · plaintext destroyed after hash
	enc_pan — Format-Preserving Encryption per Employee FF3-1 FPE(PAN, employee_DEK) · reversible · stored encrypted · never plaintext in DB
	enc_dek — AWS KMS Envelope Encryption KMS_Encrypt(DEK, tenant_KEK) · customer-managed CMK · ap-south-1 · per-employee DEK
	Document Bytes — AES-256 + Application Layer S3 SSE at rest · per-employee DEK app-layer encryption · in-memory decryption only
	Passwords Argon2id · TOTP AES-256-GCM Argon2id: time=2, memory=65536, parallelism=2 · TOTP secret encrypted separately

✳ BLAST RADIUS ISOLATION

Each employee has their own DEK. A DEK compromise exposes exactly one person's documents — not the entire tenant. Mathematics limits the damage.

🚫 ZERO PLAINTEXT POLICY

No plaintext PAN anywhere in the system after the initial 2ms HMAC window. No logs, caches, or API responses contain raw PAN or salary — ever.

🔑 AWS KMS KEY HIERARCHY

Platform secret and tenant KEKs managed by AWS KMS (ap-south-1, customer-managed). Key rotation and audit policies enforced by KMS — not application code.

🕒 PASSWORD-PROTECTED DOCUMENTS

Employee provides password → in-memory decryption only → session wiped on expiry (10 min). Nothing persisted. Zero residual data after session end.

 ALL 5 DPDP RIGHTS BUILT IN — NOT RETROFITTED

The Digital Personal Data Protection Act 2023 —

Every right exercisable in a single tap — each exercise triggers a named Temporal workflow with retry, audit trail, and SLA tracking. No manual handling. No DPO email chains.



S.11 — RIGHT TO ACCESS
Data Access & Download
See everything PRANA holds.
Download it all — one click.
Complete data package with provenance metadata. Fulfilled under 72 hours.



S.12 — CORRECTION & ERASURE
Correction + DPDP Certificate
Flag errors. Request deletion with DPDP certificate on completion. Audit log retained (7-year legal override).



S.13 — GRIEVANCE REDRESSAL
Grievance + DPB Escalation
Track grievances in-app. 48-hour auto-acknowledgement. One-click DPB escalation if unresolved at 30 days.



S.14 — NOMINATION
Family Vault Access
Nominate family for vault access.
First platform in India to implement the DPDP Act S.14 nomination right.



S.7 — WITHDRAW CONSENT
Granular Consent Toggles
Analytics toggles per purpose. Immediate effect. Withdrawal stops all processing for that purpose within the session boundary.

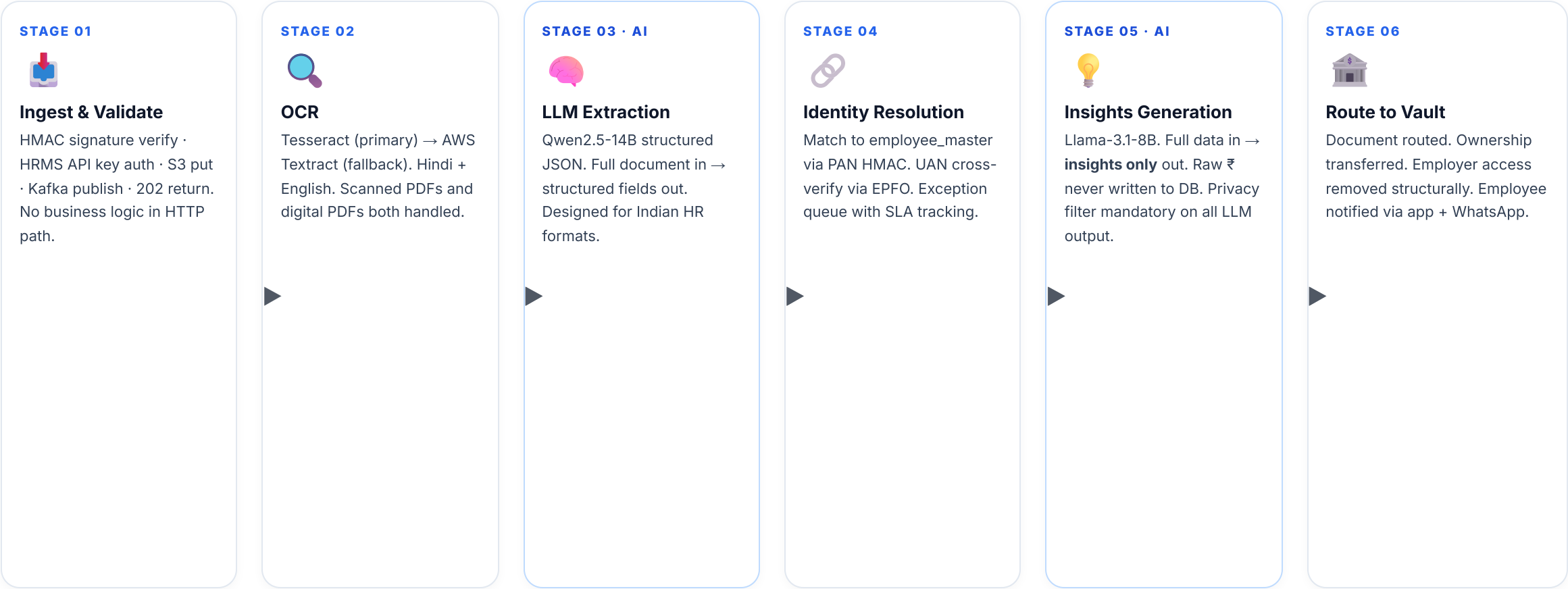
₹50Cr

Maximum DPDP Act penalty per violation. **With PRANA's architecture, misuse is structurally impossible** — not just policy-restricted. The controls are in the mathematics, not the employee handbook. Every right exercise generates an immutable Temporal workflow audit record.

 VERIFIED AT EVERY STEP · PRIVATE BY DESIGN

6-stage AI pipeline —

Every document pushed by an employer passes through a 6-stage AI pipeline. It produces insights and verification — never raw salary figures or identity numbers in any database.

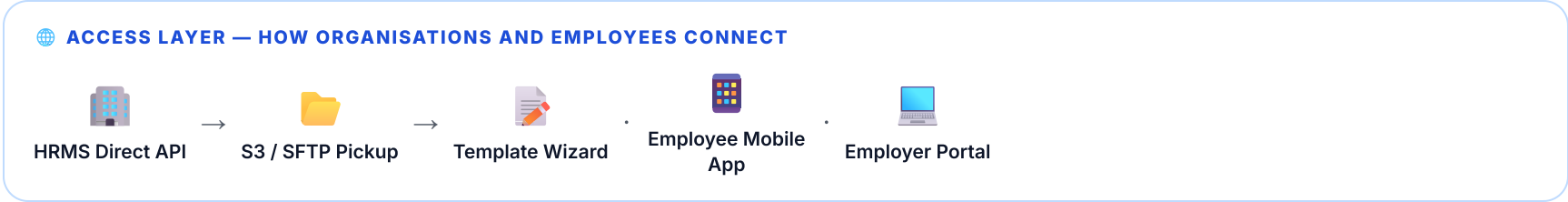


 **PRIVACY CONTRACT** **LLM Input** = full document data (needed for extraction) · **LLM Output** = insights only, no raw ₹, no PAN · **DB Write** = insights + verified flags only — never raw salary or identity numbers. Enforced in code, not policy.

⚙️ ENTERPRISE-GRADE · DUAL-REGION · EVENT-DRIVEN

Three services.

Apache Kafka event bus. Temporal for every durable process. YugabyteDB dual-region SQL. Redis CRDT sub-10ms sync.



↕ FastAPI + Kong API Gateway — 9 Service Domains ↕



↕ Persistence + Cache Layer ↕



INFRASTRUCTURE

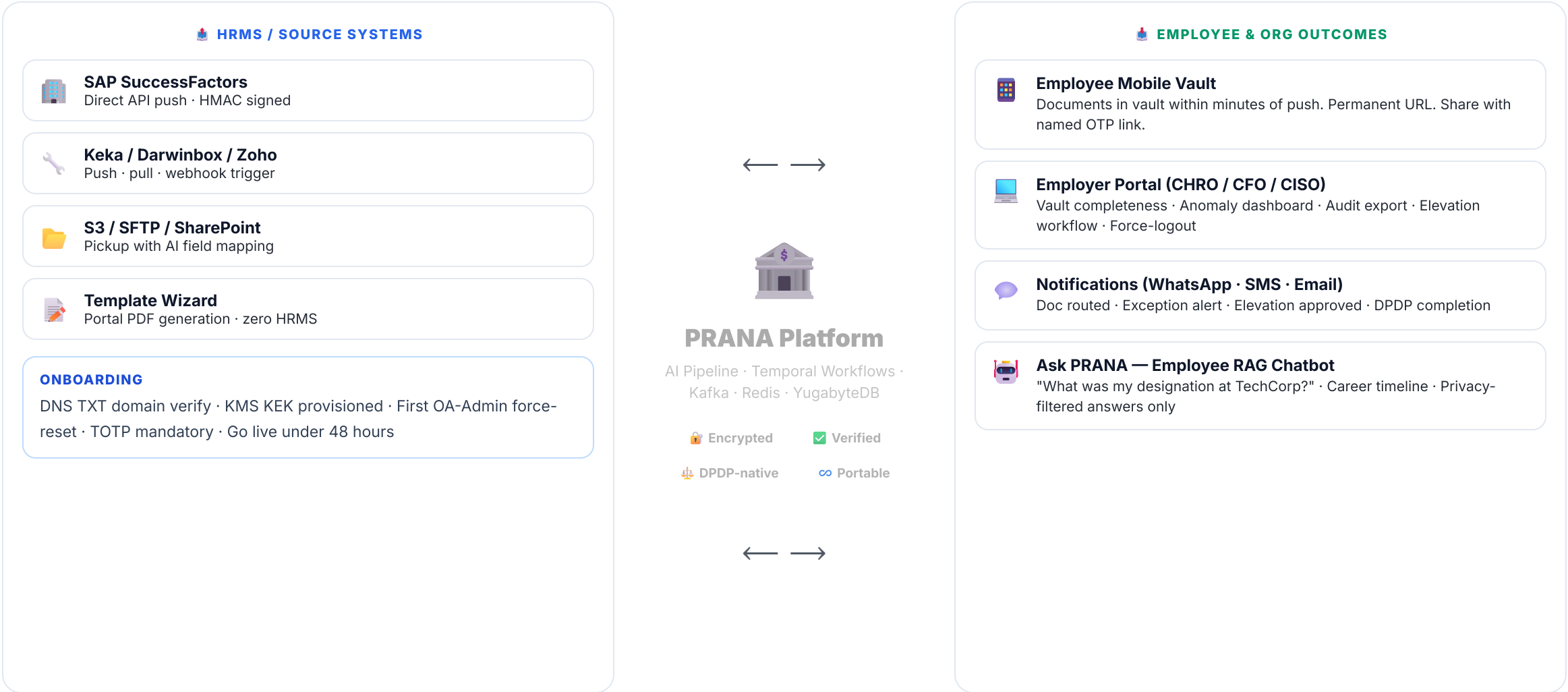
- AWS MSK · KRaft · MirrorMaker 2**
Kafka · Both regions · Bidirectional sync
- ElastiCache Global Datastore CRDT**
Redis · Active-active · Sub-10ms
- YugabyteDB · 26 tables · 256 tablets**
ap-south-1 + ap-south-2 dual-region
- AWS KMS · Customer-Managed CMK**
ap-south-1 · Tenant KEK + Platform secret
- Temporal Cloud — 53 workflows**
5 task queues · mTLS · Zero polling
- Apache Iceberg on S3**
7-year audit retention · Court-admissible
- SCALE TARGETS** 🎯
1 lakh organisations · 1 crore employees ·
125 PB S3 · Active-active dual-region

From real-time document push to employee vault in <60 seconds · Dual-region active-active · Mumbai + Hyderabad

🔑 THREE PATHS TO INTEGRATION · GO LIVE IN DAYS

PRANA meets your HRMS

Direct API push for modern HRMS. Shared location pickup for legacy systems. Template Wizard for orgs starting from zero. All paths produce the same verified, AI-extracted vault entry.



The vault is ready.

PRANA is production-ready — dual-region, encrypted, DPDP-compliant. The fastest path to employee document infrastructure and structural privacy compliance in India.



STEP 01

Technical Walkthrough

60-minute deep-dive with your InfoSec and HRMS leads. Encryption architecture, integration options, and tenant onboarding — in detail.



STEP 02

Pilot Tenant Setup

One department. One HRMS connector. Domain verification and KMS provisioning under 48 hours. First documents in vault before any enterprise agreement.



STEP 03

Enterprise Rollout

Full employee base onboarded. AI pipeline calibrated for your document types. CHRO, CFO, and CISO dashboards configured to your org structure.

🇮🇳 **Built for India.** DPDP Act 2023 from day one. Dual-region AWS Mumbai + Hyderabad. All data residency in India.

🔒 **Three guarantees:** Cryptographically Yours · Verified Not Just Stored · Follows You Forever